Report

A pre-registered *in silico* refutation programme on a redox-switch model of parvalbumin-interneuron damage

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Abstract. We built a mechanistic ODE model of oxidative damage to parvalbumin (PV) interneurons — a bistable peroxiredoxin/sulfiredoxin redox switch, a two-stage inducible defence ($\tau_m = 12\,\mathrm{h}$, $\tau_s = 63\,\mathrm{h}$) with priming, asymmetry, and per-cell capacity, a Phillips–Robinson sleep–wake switch, and a Wilson–Cowan cortex closing the loop through disinhibition — and spent most of the effort trying to break it. Every hypothesis was written as a numbered rule; every threshold was derived from expectations and variances before the computation; verdicts sit in an append-only ledger and are never revised. Of 128 pre-registered parts across 26 rules, 66 passed, 48 failed, 13 were unfalsifiable, one fell outside its own analytic bracket. The model describes the generation and termination of an episode under oxidative load; the failures are the main result.

1. Object of study

Hypothesis. PV interneurons hold cortical excitation in check, set the gamma rhythm, and pay with the highest energy turnover in the network, hence a large peroxide flux. Part of the antioxidant defence is produced only while the NMDA receptor — the cell's activity sensor — works; a weak receptor means less defence and more oxidants, and oxidants suppress the receptor: a positive loop. The overloaded PV cell does not die but *silences* (less parvalbumin, less GAD67), the cortex disinhibits, load rises: a second loop; a disinhibited hippocampus raises dopamine responsivity (Grace circuit), the bridge to positive symptoms. Each arrow is supported experimentally; the chain as a whole never has been. The quantitative question: is the loop gain large enough to *close*, i.e. to make the cell bistable? Bistability is what converts slow accumulation into a sharp transition with hysteresis — the clinical shape (long prodrome, abrupt first episode, episodes of hours–days).

Strategy. Build the minimal model; then falsify it systematically: numbered rules with parts, thresholds from analytic expectations only, append-only verdict ledger, bitwise-verified code, frozen reference implementation. Versions: v3 added the sleep switch, cortex, population; v4 the two-stage defence, priming, staircase, and the statistical discipline; v5 the asymmetric defence, the three-phase population, the fused kernel, and the two loop rules (F26–F29).

2. The model

The model is a stack of four layers with separated scales: the cell (minutes–days), the sleep switch (hours), the cortex (adiabatic), the population (via one scalar).

2.1. Redox core

Let $x \in [0,1]$ be the working fraction of peroxiredoxin, c the receptor-maintained protection, u the peroxide, ν the load. u is fast (algebraic balance); the dynamics are

$$\begin{aligned} &\nu = \varepsilon u + \frac{xu}{1+u}, \qquad \varepsilon = e_0 c^{\gamma}, \qquad N = \frac{N_0}{1+u/K_N}, \\ &\dot{x} = c^m(1-x) - \mu_0 \frac{x u^2}{1+u}, \qquad \dot{c} = \frac{1}{T}\Big(N + \frac{\beta u}{K_B+u} - c\Big). \end{aligned} \tag{1}$$

The first line and the x-equation are measured kinetics (Selvaggio et al. 2018); the c-equation is an assumption of form. Notation:

- e_0: scale of the non-enzymatic clearance εu; μ_0: scale of the two-hit hyperoxidation $\mu_0 x u^2/(1+u)$ relative to repair;
- m: exponent of the defence-amplified repair $c^m(1-x)$; γ: exponent of the bypass clearance $\varepsilon = e_0 c^{\gamma}$;
- K_N: saturation of the peroxide suppression of the receptor-maintained defence $N = N_0/(1+u/K_N)$, with $N_0 < 1$ a congenitally weak receptor;
- β, K_B: strength and half-saturation of the stress induction $\beta u/(K_B+u)$;
- T: defence-protein lifetime in repair-time units.

Per-term provenance with statuses: `latex/core_derivation.tex`.

Proposition 2.1 (Bistability). *Over-oxidised peroxiredoxin is repaired slowly, so working enzyme* decreases *with peroxide while clearance* requires *enzyme: the feedback in* (1) *produces two saddle-node folds and a hysteresis window of loads in which "defence on" and "defence off" are both stable. Measured cells lie outside the window; entering it requires repair weakened* 1.6–22*-fold — the proposed state of PV cells under weak NMDA signalling.*

Parameter sets. Nine parameters are sampled (e_0, μ_0, K_N, K_B, T log-uniform; $m \in [1,2]$, $\gamma \in [0,1]$, $N_0 \in [0.3,1]$ uniform) with $\beta = N_0 \cdot U(0,3)$ (measured induction up to 2.85×). A set is accepted iff bistable: window ratio > 1.05 and two stable states at the geometric-mean load. These accepted sets are the model's *potential patients*: only a set whose cell already sits in the bistable regime — the weakened-defence state the hypothesis postulates for PV cells under weak NMDA signalling — can develop the switch-mediated damage the model describes; everything else in the sampled space is healthy by construction and is not studied. $K = 200{,}000$ candidates $\to n_{\text{sel}} = 2400$ accepted sets (seeded, cached); the resting load sits at $q = 0.10$ of the window.

2.2. Inducible defence

Two stages. Induction signal m, then protein s, added to c:

$$\dot{m} = \frac{1}{\tau_m}\Big(\beta \frac{u}{K_B + u} - m\Big), \qquad \dot{s} = \frac{1}{\tau_s}(m - s), \qquad \tau_m = 12\,\text{h},\ \tau_s = 63\,\text{h}, \tag{2}$$

with step response $s(t) = 1 - (\tau_s e^{-t/\tau_s} - \tau_m e^{-t/\tau_m})/(\tau_s - \tau_m)$ — always sigmoidal; half-level at 57 h. The stationary state (hence window, folds, gap law) is unchanged; only the transient changes.

Priming. A trace p on the induction signal amplifies it: $\dot{p} = (u/(K_B + u) - p)/\tau_p$, $\beta_{\text{eff}} = \beta(1 + Ap)$, $A = 2$ (ceiling ×3, measured; τ_p scanned — unmeasured).

Asymmetry. Rise and decay constants of s are split, $r = \tau_\downarrow/\tau_\uparrow$.

Proposition 2.2 (Asymmetric integrator). *For a two-level daily input, high for a fraction f of the day, the stationary mean of the integrating variable stands at*

$$w(r) = \frac{fr}{fr + 1 - f}, \tag{3}$$

$f = 0.705$ (measured wake fraction). Asymmetry turns the integrator into a peak detector ($r > 1$) or trough detector ($r < 1$); no fitted numbers enter (3).

Capacity loop. Per cell, while damaged, $N_0 \to N_0(1 - a_n)$ (loss 1 d), recovery slow (30 d): a latch on recovery, not on load. *Dopamine loop* (v5.1): a tone d driven by disinhibition $E/E_r - 1$ or by D ($\tau_d = 7$ d) multiplies the load by $(1 + a_d d)$; at $a_d = 0$ the model is reproduced bitwise.

2.3. Sleep–wake switch

Phillips–Robinson, "new nominal" set (Skeldon et al. 2014), taken whole, untuned:

$$\begin{aligned} &Q(V) = \frac{Q_{\max}}{1 + e^{-(V-\theta)/\sigma}}, \quad \tau_v \dot{V}_v = \nu_{vm} Q(V_m) + D_v - V_v, \quad \tau_m \dot{V}_m = A_m + \nu_{mv} Q(V_v) - V_m, \\ &D_v = B_0 + \nu_{vc} \cos\tfrac{2\pi(t-\alpha)}{24} + A_G G - k_d D_{\text{cpl}}, \quad \dot{G} = \begin{cases} (1-G)/18.2\,\text{h} & \text{awake}, \\ -G/4.2\,\text{h} & \text{asleep}, \end{cases} \end{aligned} \tag{4}$$

wakefulness := $Q(V_m) > 1\,\text{s}^{-1}$. Here Q is the firing-rate sigmoid ($Q_{\max} = 100\,\text{s}^{-1}$, threshold θ, slope σ); V_v, V_m the sleep- and monoaminergic-nucleus voltages with mutual inhibition ν_{vm}, ν_{mv}; D_v the drive to the sleep nucleus — constant part B_0, circadian amplitude ν_{vc} with phase α, somnogen weight A_G; G the somnogens; $-k_d D_{\text{cpl}}$ the sleep–damage coupling, with D_{cpl} the mean damage or the damaged fraction. Checks against publication: folds at 1.450/2.463 mV (paper: 1.45/2.46); free cycle 15.5/8.5/24.0 h; somnogen swing 0.527 vs. 0.50 — unfitted. Damage enters via $-k_d D_{\text{cpl}}$ ($k_d = 2$ mV): a damaged population suppresses the sleep nucleus and lengthens wakefulness.

2.4. Cortex

Wilson–Cowan pair (E excitatory, I PV), $F(z, a, \theta) = 1/(1 + e^{-a(z-\theta)})$:

$$\dot{E} = -E + F(c_1 E - c_2 g I + P_e), \qquad \dot{I} = -I + F(c_3 E - c_4 g I), \qquad g = 1 - 0.4\, D, \tag{5}$$

$P_e = P_{e0} + P_{e1} \min(Q(V_m)/Q_{\text{wake}}, 1)$; here c_1–c_4 are the connectivity weights, P_e the wakefulness-borne drive to E, Q_{wake} its wake threshold. The multiplier on *all* PV output weights with $g_{\text{dep}} = 0.4$ is from Volman et al. 2011 (inhibition to 60 %). Cortical scales are 10^3 times faster than everything else: stationary solution by damped iteration (adiabatic elimination); the gamma rhythm is absent by design. At full damage E rises 19 % — a consequence of two literature numbers, not a fitted gain.

2.5. Population

$J = 32$ cells differing in one scalar, the load multiplier $v_j = \exp(\sigma_v z_{p_j})/\text{mean}$: $\sigma_v = 0.10$ is the spread and z_{p_j} the fixed normal quantiles of the edge-pinned probability grid (pinned to $J = 32$). Cells couple only through network activity (load on survivors rises with the damaged fraction). Gradualness is *from the spread, not the network* (F5, refuted). The grid subtlety: J is a model parameter through the realised spread, $\text{std}(\ln v) = 0.1226/0.1060/0.0980/0.0943$ at $J = 8/16/32/64$ vs. 0.0910 in the limit ($O(1/J)$ convergence), with log-sensitivity ≈ 2.5 (section 4.4).

2.6. Load, episodes, and the gap law

The load $L(t)$: base at $q = 0.10$ of the window; circadian swing 20 % — the weight A_H of the somnogen deviation $G - \bar{G}$ in the load, calibrated on the switch; fast noise $\sigma = 0.08$ on log-load, correlation $0.3T$; slow noise *off* (withdrawn in F6, F18); optional serotonin bursts ($\times 1.5$, rate 0.1/day, decay 2 h, ceiling $\times 2$); auxiliary protocols for the loop rules (two-phase break–release; static multiplier n_{mul} as the loop's static equivalent).

Definition 2.3 (Episode, band)**.** *An episode is the time with damaged fraction $D > 1/2$ (the continuous mean-damage variable for cross-J comparisons; crossing moment interpolated inside the step). The* band *is the set of parameter sets whose upward gap is covered by the available burst and whose downward gap fits the noise budget $2\sigma + \sigma_s$.*

Proposition 2.4 (Gap conservation)**.** *With window ratio W, load position q (log scale), feedback factor F, daily peak and trough P, T:*

$$\ln \frac{W^{1-q}}{P} + \ln(W^q F T) = \ln \frac{WFT}{P} \quad (q\text{-independent}). \tag{6}$$

Typical sets: sum $\approx 9\,\sigma$ of load noise; the clinical picture (rare episodes of hours–days) needs $\approx 4.5\,\sigma$. *The same noise cannot both break rarely and release quickly.* The exits: a one-sided trigger, or a variable defence that raises its own lower boundary — the pursued route (section 2.2).

3. Computation

3.1. Layout

File	Role
`run.py`	entry point: args, criteria self-check, blocks, report, ledger
`config/base.toml`	all run parameters ($n_{\text{sel}} = 2400$, 60 d, $M = 1600$, $J = 32$, warm 16 d, $\sigma = 0.08$, $k_d = 2.0$, $\sigma_v = 0.10$, $\tau_s = 2.625$ d)
`px/consts.py`	constants and sigmoids, sources in comments
`px/core.py`	stationary curve, window, set sampling, vulnerability grid
`px/sleep.py`	switch: folds, free cycle, calibration
`px/cortex.py`	Wilson–Cowan stationary solution
`px/model.py`	`simulate`: one step over all sets and scan blocks
`px/fast.py`	fused numba kernel (3.3×) + fast `simulate`
`px/conf.py`	TOML config, set cache, layers, gap law, band
`px/batch.py`	blocks, hash-dedup, per-block parallelism, block cache (keyed by model-source hash)
`px/stats.py`	criteria: sign test, median bound, CV^2 of gaps, MDE adjudication, self-tests
`px/predict.py`	analytic expectations before any simulation (risk model)
`px/report.py`	run report and append-only `ledger.json`
`px/rules/fNN.py`	one rule per file: `needs(cfg)`, `run(ctx)`
`px/rules/_blocks.py`, `_meas.py`	shared block constructors; per-block summary
`mass_core6.py`	frozen reference monolith, not used in compute
`tests/test_regression.py`, `tests/test_fast.py`	bitwise checks vs. monolith / kernel

3.2. Pipeline, blocks, rules

```
python run.py --list                 # available rules
python run.py --rules all --dry      # block/time estimate, no compute
python run.py --rules f19            # one rule; f02,f06,f18: several
python run.py --ledger               # accumulated verdict journal
python tests/test_regression.py      # bitwise check vs mass_core6.py
```

Only requested blocks are computed; identical blocks are deduplicated by hash (78 requested → 40 unique). Block results are cached under a key containing the model source hash: editing the model invalidates the cache by design. Parallelisation is *per block only*: noise is drawn for one block length and repeated over scan points, so blocks are strictly paired. A full run at 2400 sets ≈ 3 h (fused kernel). Rule discipline: (i) blocks through shared constructors in `_blocks.py` — same block for all rules, or pairwise comparisons fall apart; (ii) every part through `stats.adjudicate`/`sign\_adjudicate`, which return "unfalsifiable" symmetrically when the minimum detectable effect exceeds the threshold; (iii) every part records `ctx.verdict(...)` into `ledger.json` (date, block hashes) — verdicts are never revised; defects are annotated beside the verdict and re-asked as a new rule on a fresh seed.

3.3. Statistics

A criterion is usable only if its null value, its noise at our sample size, and its power are known before the run:

$$\text{MDE} = (z_{0.95} + z_{0.80})\,\text{SE} = 2.4865\,\text{SE}; \tag{7}$$

agreement claims use a 3 SE tolerance that must exceed the MDE, or the claim is unfalsifiable in principle. The sign test carries a *mandatory direction* (a silent sign error once turned $p = 0.038$ into $p = 0.989$); medians use the one-sided order-statistic bound; gaps use within-set CV^2 (null exactly one); every criterion self-tests on synthetic data with known answers before any run. Analytic expectations `px/predict.py`: the log-load sits at $m_0 = \ln(\text{lo} \cdot (\text{hi}/\text{lo})^q)$, the barrier at $\ln(\text{hi})$; an onset is a noise excursion over the barrier, rate $h(t) = e^{-d(t)^2/2}$ with $d(t) = (G - \ell(t))/\sigma$ (pre-exponential cancels in all ratios). This one model yields, unfitted: the late-wake onset share (F2a: 0.82), the swing-response curve, the mid-scale fork (F5), the burst-trigger share (F8a), the wake-lengthening ratio from the deterministic switch (F4a, F6a), the affine duration law (F7, F8b). A rank correlation on 8 cells has error $1/\sqrt{7} = 0.38$; the fix is per-set slopes with attenuation $\rho_{\text{set}} = \rho_{\text{pool}}/\sqrt{1+\lambda}$ — in the real data $\lambda = 33.8$ turned an MDE of 0.94 into 0.076 and closed four of five parts.

3.4. Reproducibility

(i) *Frozen monolith*: `tests/test_regression.py` requires the set sampling, band arithmetic, and `simulate` to match `mass_core6.py` bitwise on pinned configurations; cross-machine comparison uses 10^{-12} with a machine fingerprint (measured worst 3×10^{-15}). (ii) *Fused kernel*: assembled from the reference source by textual substitution with uniqueness checks (an edit breaks the assembly, never diverges silently); verified on all 67 output fields over nine pinned configurations (counting bitwise; worst continuous deviation 9.5×10^{-15} vs. 10^{-12}) and by diffing all verdicts; 3.3× wall-clock gain. (iii) *Step size*: continuous means converge to 0.5 % at $M = 800$; counting quantities do not — the Richardson limit is 2.0 % above the value at $M = 1600$ — so counting thresholds finer than $\approx 3\,\%$ are not trusted, stated rather than hidden.

3.5. Seven statistical traps

Each once produced a false conclusion in our hands; each is closed by a code check: (1) median over sets of a rare-event share is degenerate; (2) the share of short gaps $\equiv 1 + \mathrm{CV}^2$ of set rates — physics-free; (3) sign-test direction is a mandatory argument; (4) zero error by construction in self-normalised columns; (5) a saturating measured quantity drowns real effects — measure the rate, not the ever-share; (6) an equivalence tolerance wider than the curve's own range is unfalsifiable; (7) the mean of a bimodal population ($D > 0.9$ vs. $D < 0.01$) is the share of broken sets, not a state.

4. Results

4.1. The ledger

128 pre-registered parts across 26 rules (F2–F29): 66 passed, 48 failed, 13 unfalsifiable, 1 below its analytic bracket (P/F/U/B in the table).

Rule	Question	P/F/U/B
F2	onsets late in wakefulness; swing threshold	2/4/0/0
F3	damage delays sleep, lengthens episodes	8/0/1/0
F4	sleep deteriorates before the threshold (precursor)	3/3/1/0
F5	gradualness of the damaged fraction	0/1/0/1
F6	precursor: height, slope, slow noise	4/2/2/0
F7	duration within the two-day target	3/3/0/0
F8	serotonin burst as trigger	2/7/0/0
F10	episode composition: tail vs. time above window	3/2/0/0
F11	induction closes episodes	1/2/0/0
F12	deaf (refractory) period from defence	1/2/0/0
F13	independence of cell count J	2/2/0/0
F14	defence memory masked by noise	2/1/0/0
F15	refractory period exists, not from defence	3/0/0/0
F16	affine duration law, out-of-fit check	1/2/0/0
F18	who makes the precursor	2/2/2/0
F19	priming reaches the clinical target	4/0/0/0
F20	rhythm breaks via capture staircase	7/0/0/0
F21	third slow variable = population filling	2/1/0/0
F22	J via effective spread (substitution check)	1/1/0/0
F23	asymmetric defence follows $w(r)$	3/0/1/0
F24	bursty load beats smooth load	0/2/2/0
F25	three phases; band shape	3/2/0/0
F26	dopamine loop vs. static equivalent	2/2/0/0
F27	coexistence window of the load loop	2/3/0/0
F28	capacity loop, first edition (controls failed)	0/3/4/0
F29	capacity loop, matched differences	5/1/0/0

4.2. Established

1. *Duration is set by defence memory, not repair capacity.* 99.8 % of the within-set duration variance is explained by load history, 0.2 % by repair (exact decomposition, shares sum to 10^{-16}). The law is affine:

$$\text{duration} = 1.28 + 0.97\,\tau_b \text{ (days)}, \tag{8}$$

fitted below 100 h, checked at an out-of-fit 150 h; a shared-offset saturating form is rejected (F16b) though the quantitative out-of-fit prediction missed its 2 SE band (F16a). Passive repair misses the two-day target by 2–10× (the needed memory, ≈ 12 h, is below the first stage).

2. *Slow defence memory makes the pre-episode precursor:* removing it lowers the precursor in 107 of 174 sets, $p = 0.0015$; slow load noise is unnecessary.
3. *Three phases, not two.* A band of intact cells *more* vulnerable than their collapsed neighbours: 0.00732 ± 0.00042 of the population (17σ; ≈ 8.5 % on a fine grid); exactly zero with frozen defence; 1.82× wider under priming. Population-level hormesis, a model prediction, not a counting artefact.
4. *Priming (F19) reaches the clinical target:* median duration 1.46 d against a one-sided 95 % bound of 1.665; short episodes 0.363; paired $p = 5.2 \times 10^{-6}$; all four parts pass, replicated on 2000 sets. Unfitted: ceiling ×3 measured, τ_p scanned; accelerating the first stage gives −1 to −5 % vs. −52 % for amplification, as predicted. Mechanism: priming raises the *top* of the window, not repair speed.
5. *Asymmetric defence (F23):* scanning $r = 0.33, 0.5, 1, 2, 3$ predicts $w = 0.441, 0.544, 0.705, 0.827, 0.878$ by (3); measured defence levels $0.0575, 0.0644, 0.0792, 0.0958, 0.1065$ — order exact ($\rho = +1$, permutation $p = 0.0083$). Damaged share falls $0.486 \to 0.457$ ($p = 0.048$); duration $7.74 \to 6.88$ d. Equal rise/decay constants were a *choice of regime*, not a simplification; r must be a model parameter unless measurement fixes it.
6. *Capacity loop (F29) converts frequent short episodes into rare long ones:* duration $13.1 \to 25.6$ d while the onset rate *falls* $0.042 \to 0.033$ per at-risk set-day; episodicity 0.75 of the open model; the broken state is held after release, monotone in release depth ($\rho = +1.000$, $p = 0.0083$); progression exceeds a capacity-matched static control by +0.0231 vs. MDE 0.0096, replicated as +0.0223, +0.0211 on two further seeds. The one failing part: the *shape* of risk in time is unchanged (risk trend 1.94 vs. threshold 2.23, F29c).
7. *Rhythm: a capture staircase* with rational rotation numbers and alternating cycle lengths on fractional steps (section 4.4).
8. *J enters only as effective spread of resilience;* the $J \to \infty$ limit exists with $1/J$ corrections (section 4.4).

4.3. Refuted

- *Grace-type load loop (F26, F27):* irreversibility latches at loop strength 1.11 while episodes extinguish at 0.18 — a sixfold structural gap, robust to a gentler protocol; no window with chronicity and episodicity together. The open model releases all broken sets ($0.9936 \to 0.0000$), the loop releases 0.24 %. Reason: load is per-set, so the loop moves the whole population.
- *Capacity loop:* level and duration change, the shape of risk in time does not (above).
- *Bursty load does not beat smooth load (F24):* onset rate falls 40 % as correlation time grows at fixed mean and variance; the ever-damaged share saturates.
- *Serotonin burst (F8):* 20 % of onsets follow a burst vs. 80 % required — a volume knob, not a trigger; no episode property changes.
- *Daily swing (F2b):* smooth 1.5× growth, no gate; gradualness is from spread, not network (F5, observation *below* the analytic fork).
- *The episode does not remember its depth (F10a):* duration matches time-above-window (rank correlation 0.94 old / 0.87 current model); tail ≈ 7 % of the episode.
- *Band shape* $\varphi(\Phi^{-1}(1 - D))$ refuted twice with opposite signs (F25g,d) — width is not a function of D.
- *J-substitution check (F22b) failed:* the effective spread explains the measured drop but does not close the question (candidates: $1/J$ mean-dispersion, discreteness of D in the sleep feedback).
- Several precursor-slope parts were unfalsifiable at the chosen grid resolution (MDE larger than the written threshold).

4.4. Selected mechanisms

The staircase (F20). The deterministic switch under 24 h forcing, with damage inhibiting the sleep nucleus: the rotation number is pinned at rationals and jumps:

$$\begin{aligned} 1{:}1 \to 5{:}4 \to 3{:}2 \to 2{:}1 \to 3{:}1 \to 4{:}1 \to \cdots \to 9{:}1 \\ \to \text{no sleep}, \end{aligned} \tag{9}$$

transitions at $kD \approx 4.77$; 4.78; 4.79; 5.09; 5.16 mV; five steps (5:1–9:1) within 5.172–5.178 mV. A fine run: 52 steps in 0.9 mV, 34 with alternating cycle lengths (patterns $25, 24, 24, 47$ h at 4.772; $47, 25$ at 4.779; $46, 26, 24$ at 4.783); widths 0.3 mV (2:1), 8×10^{-3} (3:2), 5×10^{-6} (11:8). Capture staircase (Arnold tongues), *not* a period-doubling cascade: the period adds one skipped night; the discriminator is a capture on a non-power-of-two period (found). Limitation is calibration: $kD \leq k$ identically, so at $k = 2$ mV the first step (4.77 mV) is unreachable at any damage; at $k = 5$ mV the model reaches it ($kD = 4.93$, $D = 0.999$).

Cell count J (F13, F22). Three paths: (1) the tail of the vulnerability distribution — closed by the fixed-edge grid (at $J = 32$ identical to the historical grid, last digit); (2) quantisation of the episode threshold (0.625 at $J = 8$ vs. 0.516 at $J = 64$) — bypassed via the continuous variable; (3) the realised spread (section 2.5): 11 % in effective σ_v between $J = 16$ and 64, at log-sensitivity $\approx 2.5 \Rightarrow \approx 25$ % in measured quantities — exactly the range F13 found at 2000 sets. A discretisation can preserve mean and edges exactly and still miss the variance ($O(1/J)$): comparing runs at different J compares different models.

The capacity loop (F28, F29). The first edition collapsed on its own controls: "no increment with the loop open" is false — a bistable switch never releases a broken set, so the broken share rises on its own; the control passed at small n and failed as data accumulated. F29 therefore contains *no test against zero*: all claims are matched differences at equal damage level (section 5).

The band (F25). On a 600-node grid: frozen defence $\Rightarrow$ exact monotonicity (zero violations); live defence breaks order on 11.9 % of steps, worst violation 0.973 (full reversal); priming amplifies, does not cause ($A = 0$: 0.0184). Consequence: the closed-form damaged fraction $1 - F_v(v^*)$ is illegal under live defence; any reduction needs an explicit three-configuration boundary layer.

5. Methodological lessons

In a model with absorbing states and a heterogeneous population, *no absolute null exists for any accumulation measure*: the damaged fraction rises on its own (bistable sets do not release) and the risk falls on its own (the vulnerable are absorbed first). Two of our controls asserted these baselines were zero; both were false and both *passed at small sample and failed as data accumulated* — a false null is passed by lack of power, since the threshold shrinks as $1/\sqrt{n}$ while the baseline is fixed. Every claim must be a matched difference between arms at the same damage level. The seven traps (section 3.5) are this lesson in seven costumes; one more: the range of n noisy values is itself random ($\mathbb{E}[\text{range}] = 1.693\,r$ for $n = 3$), so a "spread below 10 %" threshold was meaningless at $r = 12$ % and became testable only at 2000 sets — where it failed and exposed the third path of J.

6. Scope and open questions

The model has no absorption-free stationary state: it is valid in windows short compared with the accumulation time — a model of the *generation and termination of an episode*, not of the course of illness. The capacity loop is the only mechanism tested that makes progression appear, and it does so by lengthening episodes, not by multiplying them. Open: (i) progression on illness scale needs a component that does not fully return — deliberately not added: a mechanism introduced to produce its target result is not a test; the falsifying rule comes first; (ii) regions coupled through narrow bottlenecks vs. the per-set homogeneity that killed the load loop; (iii) rhythm breakdown unreachable at $k = 2$ mV vs. first step 4.77 mV — miscalibration or another path; (iv) the band width is set by something third (likely the time near the tipping point) — testable against the mid-scale residence time, must agree on

both scans or it is withdrawn; (v) the third slow variable behind the precursor (population filling has no own τ); (vi) the free term of (8) (1.3 d at zero memory) is unexplained by x; (vii) peroxiredoxin kinetics are from cell lines, not PV cells; (viii) dopamine and symptoms are outside the model by design.

7. Reproducibility

Commands and configuration: section 3. Verdicts: `ledger.json` with dates and block hashes. Reference: `mass_core6.py` (frozen); bitwise checks: `tests/`. Full run at 2400 sets $\approx$ 3 h; always `-dry` first. The model-source hash enters the block-cache key: editing the model invalidates the cache by design.

Acknowledgements. Text and simulation code were produced with Claude Opus 5 and DeepSeek, as in the companion paper `paper.tex`; all numbers come from the project's own ledger, reports, and source comments; biological anchors carry provenance marks in the brochure `brochure.tex`.

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